

Saejoon Park

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EDUCATION

Cornell University | Ithaca, NY

Aug 2024 – May 2028

Bachelor of Science in Electrical and Computer Engineering (Robotics Concentration)

SKILLS

Programming Languages: Python, C, C++, Swift

Software/Platforms: KiCad, LTSpice, MuJoCo, Fusion360, Git/GitHub, Claude Code

Languages: English (fluent), Korean (fluent)

EXPERIENCE

Wim Robotics | Summer Research Collaborator

May 2026 – Aug 2026

- Designing an **Analog Neural Network**, a learned mixed-signal actuator controller: a continuous-time recurrent neural network (3 inputs, 4 hidden states, 1 torque output) in analog hardware computing nonlinear inverse dynamics as current-mode feedforward, closed by a digital ADRC loop with an extended state observer, targeting high-precision control inside tight SWaP budgets.
- Laid out the full KiCad schematic and PCB: 64 digital potentiometers across the 4×3 input and 4×4 recurrent crossbars, signed weight pairs about a mid-supply rail, op-amp σ and integrator blocks, and 8 daisy-chained 74HC595 shift registers driving every weight from 3 RP2040 pins on one latch edge.
- Built a **UR5e** control stack in MuJoCo: torque-level PID joint control tuned by an ITAE gain sweep, trapezoidal-velocity trajectory tracking with feedforward, and Jacobian-based inverse kinematics driving the end effector to a target pose to grasp an object.
- Worked closely with the **CEO and CTO** on control-architecture and hardware design reviews.

IBX, Inc | Co-Founder / COO | Ithaca, NY

Mar 2025 – Present

- **Oddity 1 Chrome Extension:** used the chenglou/pretext JS library, the Google Docs API, and HEX color tracking to enable on-screen GDocs annotations (previously impossible without a Google whitelist), solving `<canvas>` rendering difficulties.
- Developed a “Cursor for Writing” system in Google Docs from a clean-room rewrite of open-source Claude Code.
- Frontend Developer / Outreach: working with the FCPS CTIO and Director of Counseling for district app approval.

Cornell Machine Poetics Research | Advisor: Sang Leigh | Ithaca, NY

Nov 2025 – Apr 2026

- **Political AI Camera:** developed Raspberry Pi hardware around a 12.1 MP sensor with software built on the Picamera2 library.
- Calls the OpenAI API (`gpt-image-1`) prompted to re-render a captured image from differing political perspectives.

Cornell Assistive Technologies Project Team | Ithaca, NY

Feb 2025 – Apr 2026

- **Sensory Overload Wearable:** developing a custom PCB for a lap wearable accommodating individuals with sensory processing disorder.

PROJECTS

Computer Vision + Glove-Based Tracking Computer Mouse | Ithaca, NY

Jan 2025

- ATmega32U4, custom conductive inputs, and an IMU assembled into a position-tracking glove device (Valorant compatible).
- OpenCV hand-gesture tracking (simulated clicks / volume control) at 30 fps.

Digital MIDI Trumpet | Ithaca, NY

July 2026 – Present

- Built an electronic MIDI trumpet around an RP2040 microcontroller: three valve buttons, capacitive touch pads for breath, and a motion sensor that shifts the pitch as the player raises or lowers the bell.
- Designed all three circuit boards in KiCad, shrinking the touch sensor board by 59%.

Thermal Electric Generator Powered Mouse | Fairfax, VA

Aug 2023

- Patent-pending hand thermal energy-powered computer mouse.
- Implemented TEG modules and a transformer to convert a 20 mV output from TEG to 3.3 V.

Audio Signal Decoding Circuit & Filter Design | Ithaca, NY

Apr 2026

- Designed and built a 6th-order MFB bandpass filter to extract encoded audio secrets; ran automated frequency-response sweeps and safely eliminated multi-volt DC offsets.

Steel Electric Bike from Scratch | Burke, VA

May 2025 – Aug 2025

- Custom-cut and assembled steel chassis; 1000 W motor; 50 V accumulator.

HONORS

USPTO Provisional Patent for TEG Hot Swap Computer Mouse

Fall 2023